

Notes on Eigenvector Continuation and Resonance Theory

Keanan Scarbro

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1 Background

1.1 Separation of center of mass

1.1.1 Deriving the TISE from the TDSE

The time evolution operator obeys the time-dependent Schrödinger (TDSE):

$$i\hbar \frac{\partial}{\partial t} \hat{U}(t, t_0) = \hat{H} \hat{U}(t, t_0). \quad (1)$$

In the cases where the Hamiltonian is time-independent, as is our case,

$$\hat{U}(t, t_0) = \exp \left(\frac{-i\hat{H}(t - t_0)}{\hbar} \right). \quad (2)$$

Acting on an energy eigenket $|E\rangle$ such that $\hat{H} |E\rangle = E |E\rangle$,

$$|E\rangle_t = \hat{U}(t, t_0) |E\rangle = \exp \left(-\frac{i}{\hbar} E(t - t_0) \right) |E\rangle_{t_0}.$$

Such a state only varies in phase as time progresses. It remains an eigenket of the Hamiltonian with energy E for all time. Using completeness of the energy eigenkets, any ket can be written as $|\Psi\rangle = \sum_E |E\rangle \langle E|\Psi\rangle$, and so the time dependence of any arbitrary state is:

$$|\Psi\rangle_t = \sum_E \exp \left(-\frac{i}{\hbar} E(t - t_0) \right) \langle E|\Psi\rangle_{t_0} |E\rangle.$$

By applying a ket at t_0 on the right of both sides of eq. (1), we can derive the more conventional form of the Time-dependent Schrödinger equation (TDSE):

$$i\hbar \partial_t |\Psi\rangle_t = \hat{H} |\Psi\rangle_t. \quad (3)$$

Now, we restrict eq. (3) to a single energy eigenspace. In that eigenspace, the eigenket evolves as $|E\rangle_t = \phi(t) |E\rangle_{t_0}$, where $\phi(t) = \exp(-iE(t - t_0)/\hbar)$ as discussed earlier. Thus,

$$\begin{aligned} i\hbar \partial_t (\phi(t) |E\rangle_{t_0}) &= \hat{H} (\phi(t) |E\rangle_{t_0}) \\ i\hbar (-iE/\hbar) \phi(t) |E\rangle_{t_0} &= \phi(t) \hat{H} |E\rangle_{t_0} \\ E \phi(t) |E\rangle_{t_0} &= \phi(t) \hat{H} |E\rangle_{t_0} \\ E |E\rangle_{t_0} &= \hat{H} |E\rangle_{t_0}. \end{aligned}$$

This approach derives the time-independent Schrödinger equation (TISE) from abstract operator algebra, which is more general than the standard approach: starting with the TDSE and using the position basis, the assumption that H doesn't depend on t , and separation of variables to arrive at the same equation.

1.1.2 Separating out the center of mass

Consider a two-particle system where the particles have masses m_1 and m_2 and the particles' positions are $\mathbf{r}_1$ and $\mathbf{r}_2$. We can define the center of mass and relative coordinate as

$$\mathbf{R} = \frac{m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2}{m_1 + m_2} \quad \mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2,$$

respectively. If the potential is central, one can then show that the Hamiltonian factorizes and the center of mass motion is independent of the relative motion. The center of mass just contributes an overall free motion and can be neglected (moving to the center of mass frame).

1.1.3 Separating out angular motion

From this, we can then separate out angular motion using the definition of the angular momentum operator. The potential is central, so the Hamiltonian commutes with the angular momentum operator. This means that there are simultaneous eigenkets of both. Angular momentum eigenstates are spherical harmonics and are degenerate. Diagonalizing the radial Hamiltonian in the degenerate eigenspace of angular momentum l , we get the reduced radial equation:

$$-\frac{\hbar^2}{2\mu} \frac{d^2 u}{dr^2} + \left(V(r) + \frac{\hbar^2 l(l+1)}{2\mu r^2} \right) = Eu, \quad (4)$$

where $\mu = (m_1 m_2)/(m_1 + m_2)$ is the reduced mass and $R(r) = u(r)/r$ is the radial wave function.

1.2 Resonance Theory

A resonance in nuclear theory is similar to a resonance in other areas of physics. Fundamentally, a resonance is a state that is self-generating but not self-contained. I.e., a resonance is a preferred state of the system, like a bound state, but it slowly leaks into the continuum. Like a bound state, the resonance needs no asymptotic incoming behavior to exist; the system exhibits the resonant behavior without needing an external crutch. But, unlike a bound state, a resonance is not permanently stable. Of course, in practice, one would need to give the system an initial kick to actually observe the resonance, but the resulting temporary state is independent of the kick's behavior. The kick is just there to give the system enough energy to transition. This is exactly analogous to resonances in other areas of physics. In classical mechanics, a bell or guitar string oscillates at a resonance frequency. It does not need to be driven at this frequency; it prefers to oscillate at it. It's a property of the system, independent of the incoming behavior. However, it can't maintain this oscillation indefinitely; it inevitably decays. Further, the bell or guitar string, practically, needs to be given a kick to slide into its resonance (the low E on a guitar string vibrates at a low E regardless of how you pluck it, asymptotically).

In fact, the math is identical as well. As discussed below, a resonance in nuclear theory is often defined as a pole of the S -matrix in a specific part of the complex plane. This defines the "self-generating but not self-sustaining behavior". In classical mechanics, the

bell's resonant frequency is a pole of its transfer function with a complex energy. The math is identical. A pole of the S matrix located on the negative real axis corresponds to a bound state: a state that is both self-generating and self-sustaining (a bound state is stable and doesn't leak). If an ideal bell were in a vacuum, its resonant state would collapse to its bound state. But, on Earth, the bell is always coupled to the air surrounding it; the bell is intrinsically open and always leaking energy to the air, so the state decays. In nuclear physics, the resonance is defined as a state that is self-generating but is open in that it leaks probability to the continuum. The analogy is nearly one-to-one.

There are two pictures for a resonance that one may use:

S -matrix picture: A resonance is defined as a pole of the S -matrix in the second Riemann sheet (identically, a pole of the analytically continued resolvent on the second Riemann sheet).

The S matrix implicitly captures asymptotic behavior. It doesn't encode the interior directly (though of course the interior governs the asymptotics). This statement is directly saying: a resonance is a state that is a pole of the resolvent like a bound state (continuum states are the branch cut), but one that is purely outgoing while not being a bound state. The resonance pole is not itself observable; it's an off-shell singularity that shapes the behavior of the continuum states near the resonance energy. This statement encodes the idea of the **Wigner time delay**: the continuum states near the resonance energy are delayed relative to free propagation (at energies close to the resonance energy, the state enters the resonance, and leaks back into the continuum, so continuum states with energies near the resonance are delayed compared to what they'd be without an interaction). The closer the pole is to the real axis, the longer this delay, and the sharper the resonance in cross-sections.

Gamow picture: A Gamow state (a resonance), is defined by purely outgoing boundary conditions with an energy that isn't pure negative real (purely outgoing boundary, but not a bound state).

If one looks at the reduced radial Schrödinger equation with this requirement in mind (by looking at the general solution for a given angular momentum l and what the asymptotic parts must then obey), one will find that this is only possible if the energy lies in the third or fourth quadrant of the complex plane (as such, we've analytically continued the solution set to the complex plane). The third quadrant solution grows in time and is called the anti-resonance, while the fourth quadrant solution decays in time and is called the resonance. This immediately means that the resulting state isn't normalizable (because of the positive real part of the energy), which is due to the divergent behavior at large r , and that the resulting state decays (because of the negative imaginary part of the energy⁺). Intuitively, this makes sense. The state must diverge because, over time, more of the state leaks out to the continuum. If you forcefully look at this state as if it were time-independent, you're collapsing all the time-dependent "leaking information" into one snapshot representing all time, so the state must diverge at large r .

⁺ It should be noted that a lot of the intuitive arguments made here implicitly assume a narrow resonance, but the concept is more general.

These two pictures are saying the same thing, and are often identical pictures (especially in the cases considered in these notes), but S -matrix picture is more general and abstract, while the Gamow picture collapses to a simplified, specific representation. In simple systems, these are identical statements (the S matrix is making a statement about the asymptotic behavior of the state, as is the Gamow condition), but a Gamow state is more difficult to define explicitly in more complicated systems.

Experiments measure the amount of particles hitting a detector for a given energy of incoming particles. The experiment effectively measures the differential cross-sections: the angular distribution of outgoing particles. Resonances show up as peaks near the real part of the resonance energy in cross sections because, at energies around that value, the particle is much more likely to be captured (to form a resonance). Once that resonance leaks out, it most likely is released isotropically, while particles that didn't interact move "straight through" and are released at $\theta = 0$. So, the cross section spikes near the resonance because, near those energies, far more particles are released at $\theta \neq 0$. It should be noted that cross-sections can't directly measure bound states. If the state is bound, it doesn't leak, so there's no particle that leaves to hit the detector. I.e., bound states occur at negative energy, and so they can't show up in scattering experiments directly at all since you can't scatter at negative energy. One must use another technique, like spectroscopy, to find bound states. Lastly, it should be noted that the theoretical treatment finds the resonant modes of the full Hamiltonian in isolation, and the scattering experiment is then one way to excite and detect those modes. The modes exist outside of the experiment, but it's how we access them. Example: $^{239}\text{U}^*$ is a highly excited resonance state formed from a neutron and a ^{238}U nucleus. The $n + ^{238}\text{U}$ is the composite system analyzed in isolation, but experiments access this by firing neutrons of different energies at a ^{238}U nucleus.

A lot of the complexity in describing resonances is that we're collapsing a time-dependent phenomena into a time-independent framework. The resonance itself is time-independent. It's a static property of the Hamiltonian. But, the realization of this resonance in any given experiment is purely time-dependent: it leaks out "over time". The pole is the state-independent object; the resulting state is what happens when you happen to excite it. We're hiding the time-dependence away, similarly to how one hides high-energy physics in effective theories.

1.3 Complex Scaling and the ABC Theorem

In complex scaling, one can show that making the substitution $r \rightarrow re^{i\theta}$ can make resonances appear bound. This is analytically continuing the Gamow state to the complex plane and evaluating it on a complex r -ray. Because of the $e^{i\theta}$, this can damp the diverging oscillations (for suitable choice of θ) and make the Gamow state L^2 normalizable.

The ABC theorem says that: if the potential is dilation-analytic (the analytic continuation of the potential is analytic in a specific θ wedge), then the complex scaled Hamiltonian is analytic in that wedge, the continuum is rotated to the position of the ray, and resonances become bound states (in that specific θ wedge) with the same resonance energy as before. This has huge computational advantages because it allows us to find resonances through

bound state methods, and is usually easy to implement computationally.

Because the complex scaled Hamiltonian is no longer normalizable, the Hermitian inner product is no longer a feasible definition for the inner product (one can normalize according to any convention one wants, and you can certainly recover the same physics if you continue to use the Hermitian inner product, but you have to be extremely careful to recognize assumptions you'll probably naturally make even though they no longer hold). Mainly, eigenvectors of the Hamiltonian are no longer naturally orthonormal wrt the Hermitian inner product, which makes insertions of identity unfortunate to work with. Defining an inner product as the c-product fixes these issues: left and right eigenvectors become identical again, the dual space structure naturally implies the left-eigenvector structure, and eigenvectors of the Hamiltonian are once again orthonormal).

1.3.1 Proof that most observables are ϕ independent

One can further show that, for any observable A that is dilation-analytic and doesn't blow up pathologically (falls off faster than a specific exponential rise), expectation values like $\langle A_\theta \rangle$ are θ independent. I.e., the expectation values are regularized observables. I'll put this proof here shortly. The general idea is to show that $\langle A_{\theta+\delta\theta} \rangle = \langle A_\theta \rangle$ (where θ and $\theta + \delta\theta$ are in the good sector) by showing that the integrals end up being equivalent because A is dilation analytic in the wedge, so there is no residue accumulated by integrating over the $\delta\theta$ ray.

2 Eigenvector Continuation

Eigenvector continuation is a reduced basis technique that allows us to find the eigenvectors and eigenvalues of a Hamiltonian that depends on a parameter λ at a target λ_* by projecting the Hamiltonian at λ_* onto a subspace spanned by eigenvectors of the Hamiltonian at neighboring values of λ . This can reduce the size of the effective Hamiltonian significantly, making the problem more computationally tractable.

2.1 Motivation of Eigenvector Continuation

- Suppose we have a parameter-dependent Hamiltonian:

$$H(c) = H_0 + cH_1$$

- In practice, direct diagonalization of $H(c^*)$ at the target value c^* may be too expensive (e.g. large lattice).

2.2 Eigenvector Continuation (EC): Core Idea

- Explicitly diagonalize $H(c_i)$ at several tractable sample points $\{c_i\}$.
- Collect the corresponding eigenvectors $\{|v_i\rangle\}$.

- Approximate the target state $|v(c^*)\rangle$ as a linear combination:

$$|v(c^*)\rangle \approx \sum_i c_i |v_i\rangle$$

- Project $H(c^*)$ onto the subspace spanned by $\{|v_i\rangle\}$:

$$\sum_j \langle v_i | H(c^*) | v_j \rangle c_j = E \sum_j \langle v_i | v_j \rangle c_j$$

- Solve the generalized eigenvalue problem:

$$H_{\text{proj}} c = E S c$$

- This only assumes that the eigenvectors at the target parameter c lie in the span of the sampled eigenvectors.
- If $H(c)$ is analytic in c , its eigenvalues and eigenvectors are also analytic (from perturbation theory), so locally: $|\psi(c)\rangle = |\psi(c_0)\rangle + \left. \frac{d|\psi\rangle}{dc} \right|_{c_0} (c - c_0) + \dots$
- Approximating the derivative via finite differences between neighboring points:

$$\left. \frac{d|\psi\rangle}{dc} \right|_{c_i} \approx \frac{|\psi(c_{i+1})\rangle - |\psi(c_i)\rangle}{c_{i+1} - c_i}$$

- Interpretation: $|\psi(c)\rangle$ is approximately a linear combination of eigenvectors at neighboring sampled points.
- Hence, as c varies, low-lying eigenvectors trace out only a small subspace of Hilbert space. with coefficients α_i determined by interpolation between neighboring points.
- Interpretation: the low-lying eigenvectors explore only a small portion of Hilbert space as c varies; they trace out a low-dimensional subspace.

2.2.1 Computational Gain

- If the full Hamiltonian acts on a space of dimension 10^6 (e.g., 100^3 lattice points), EC reduces the problem to diagonalizing a much smaller matrix—say 100×100 .
- By solving a few smaller problems (for several c_i), we can predict eigenvalues and eigenvectors across the full parameter range of $H(c)$.
- EC thus provides a non-perturbative interpolation scheme for eigenstates of large Hamiltonians.

2.3 Finite Volume Eigenvector Continuation (FVEC)

2.3.1 Extending EC to Finite Volume

- Consider simulations in finite periodic boxes of volumes $\{V_i\}$.
- Eigenvalues and eigenvectors vary smoothly with volume, encoding information about the infinite-volume limit.
- FVEC generalizes EC by treating the parameter as the box volume V instead of c .

2.3.2 Technical Subtleties of FVEC

- The inner product depends on V :

$$\langle v_i | v_j \rangle_V = \int_V d^3x v_i^*(x) v_j(x)$$

so we cannot directly mix vectors from different volumes.

- Define an enlarged Hilbert space:

$$\mathcal{H} = \bigoplus_i \mathcal{H}_{V_i}$$

- States are dilated to a common reference volume V_* , allowing projections like

$$\langle v_i | H(V^*) | v_j \rangle_*,$$

where the subscript $*$ denotes the chosen common inner product.

- The EC algorithm then proceeds identically.

3 Parametric Matrix Models

A research framework for learning parametric eigenvalue structures from sampled data. The Parametric Matrix Model (PMM) provides a way to approximate the spectral behavior of a physical system whose Hamiltonian depends smoothly on a control parameter (e.g. system size, coupling constant, or external field strength).

3.1 Overview

In many physical systems, the eigenvalues of a Hamiltonian $H(\lambda)$ evolve continuously with respect to a tunable parameter λ . Computing spectra across a dense grid of parameter values can be computationally expensive. The **Parametric Matrix Model (PMM)** aims to learn an approximate functional form of $H(\lambda)$ from a small number of samples.

PMMS were originally motivated by another computational technique: **eigenvector continuation (EC)**. In EC, one faces the same problem: explicitly diagonalizing the Hamiltonian

– which is often massive – for a dense grid of parameter values is too expensive. Instead, one diagonalizes the Hamiltonian at a few parameter values (sample points), and constructs a subspace from the eigenvectors at those points: $M = \text{span}(\{v_i\}_i) \subset \mathcal{H}$. Then, one projects the exact Hamiltonian at other parameter values onto this subspace to obtain a generalized eigenvalue problem: $H_{\text{proj}}v' = ESv'$ where

$$(H_{\text{proj}})_{ij} = \langle v_i | H | v_j \rangle, S_{ij} = \langle v_i | v_j \rangle,$$

and v' is a coordinate vector in the basis of the sample eigenvectors. This generalized eigenvalue problem has a much smaller dimension (its dimension is just the number of sample vectors taken), and is much easier to solve, than the parent eigenvalue problem.

This approach assumes that all of the eigenvectors at all parameter values of interest lie (approximately) in the span of the sample eigenvectors. This assumption is reasonable if one is interested only in the lowest few eigenstates: when H is analytic in λ , so are its eigenvalues and eigenvectors. Expanding an eigenvector in a Taylor series about some reference point λ_0 ,

$$\psi'(\lambda) = \sum_n (\lambda - \lambda_0)^n \frac{\psi^{(n)}(\lambda_0)}{n!}$$

shows that eigenvectors at nearby parameter values differ only by higher-order derivatives of ψ . In finite difference terms, this means that $\psi(\lambda)$ can be well approximated as a linear combination of eigenvectors evaluated at a few nearby parameter points. In other words, the first few eigenvectors only explore a small, local subset of Hilbert space as you vary the parameter. Projecting on this subspace allows us to solve a lower dimensional problem than the parent problem

In PMMs, one does not explicitly project the parent Hamiltonian onto a reduced subspace. Instead, the "projected Hamiltonian" is learned directly. Given training data consisting of eigenpairs at sampled parameter values,

$$H(\lambda_i)\psi_n^{(i)} = E_n^{(i)}\psi_n^{(i)},$$

the most basic PMM posits a parameterized form

$$H_\theta(\lambda) = A_\theta + \lambda B_\theta,$$

and learns A_θ, B_θ by minimizing the difference between the model's eigenvalues and the sampled true energies.

When the underlying Hamiltonian depends affinely on λ , this is equivalent to training (rather than explicitly computing) the projected matrix used in EC. However, even if the parent Hamiltonian has a more complicated or unknown dependence on λ , this approach remains valid: it effectively learns a low-order Taylor expansion of the Hamiltonian in parameter space. Additional terms (e.g., $H_\theta(\lambda) = A + \lambda B + \lambda^2 C + \dots$) or alternate functional forms (e.g., $A + 1/\lambda B + \dots$) can be incorporated to capture more complex behavior. Once trained, the PMM can predict spectra at parameter values it hasn't seen before. Interpolation between known values has already been demonstrated, but here we investigate whether PMMs can also extrapolate beyond the training domain. While PMMs are far more general than this brief description suggests, the above summarizes their role in the present work.

3.1.1 Differences between EC and PMMs

- In EC, the projected Hamiltonian H_{proj} acts as an effective Hamiltonian in the sample subspace.
- PMMs generalize this: instead of explicitly constructing $\{|v_i\rangle\}$, we train an effective matrix that reproduces the same local behavior of $H(c)$.
- We use gradient descent to learn matrices A, B in $H(c) = A + cB$ that minimizes some loss function: in our case, the mean squared error between sample eigenvalues predicted eigenvalues.
- The trained matrix then serves as a surrogate Hamiltonian in the relevant parameter region.
- Further, $H(c)$ doesn't need to have an affine dependence. We're effectively learning a low-order expansion of H around c .
- The full Hilbert space is often infinite-dimensional and contains much more information than needed.
- PMMs aim to isolate and learn the “active” region of Hilbert space relevant to the dynamics.
- Once trained, the PMM allows efficient prediction of eigenvalues and observables in that restricted region—without constructing or storing the full basis.

4 Discrete Variable Representation

To be added.

5 Density, Radius, and Regularization Methods

5.1 Radial Density

Radial density defined as $\rho(r) = \langle \Psi(r') | \delta(r - r') | \Psi(r') \rangle$. Let $\hat{\rho}(r) = \delta(r - r')$. In our case the density comes out to be $\rho(r) = \Gamma u(r)^2$. In more complicated systems, the result is more complicated.

5.1.1 Complex Scaling the Density Operator

The $\delta(r - r')$ is not something that can be complex-rotated in a traditional sense. It's a distribution, and it definitely doesn't behave in the well-defined way that most observables do (the conditions outlined in section 1.3.1. As such, there's not really a well-defined way to define the rotated version of $\delta(r - r')$. Instead, we can define an operator $\rho_\theta(r) = (\psi_\theta | \delta(r - r') | \psi_\theta)$. I.e., we don't attempt to rotate the Dirac delta. This expression is of course θ -dependent, and so the goal would be to compute it at $\theta = 0$. As such, we need to involve back-rotation.

In our case, $\rho_\theta(r) = (\psi_\theta|\delta(r-r')|\psi_\theta) = \Gamma e^{i\theta} u_\theta(r)^2$. This varies with θ , but the interior stays nearly constant. The $e^{i\theta}$ factor comes from the $e^{3i\theta/2}$ factor in the 3D complex scaling operation $\Psi_\theta(r) = U_\theta \Psi(r) = e^{i3\theta/2} \Psi(re^{i\theta})$.

5.1.2 Finding the density through backrotation

The back-rotated density is just $\rho^{\text{back}}(r) = \Gamma u_0(r)^2 = \Gamma u_\theta(re^{-i\theta})^2$ because the back-rotated wavefunction is just the original Gamow state. Casting this with respect to the psuedo-rotated density:

$$\rho_{\theta,BR}(r) = e^{-i\theta} \rho_\theta(re^{-i\theta}),$$

One can alternatively show this with a similar integral argument that justified ρ_θ by back-rotating the resonance wavefunction: $\Psi_{\theta,BR}(r) = e^{-i3\theta/2} \frac{u_\theta(e^{-i\theta}r)}{e^{-i\theta}r}$ to recover the Gamow state, then computing the density with the standard delta function.

5.2 Radius squared operator (r^2)

5.2.1 Rotated radius squared operator

$\langle r^2 \rangle_\phi$ is radius squared operator of CS'd state ψ_ϕ . $r \rightarrow e^{i\phi}r$, so

$$\begin{aligned} \Psi_\theta(r) &= U_\theta \Psi(r) = e^{3i\theta/2} \Psi(re^{i\theta}) \\ &= e^{i\theta/2} \frac{u(re^{i\theta})}{r} Y_0^0. \end{aligned}$$

The $e^{3i\theta/2}$ factor comes from requiring that the complex scaling operator is unitary in 3D.

$$\begin{aligned} \langle r_\theta^2 \rangle &= \frac{1}{4} \frac{(\Psi_\theta | r_\theta^2 | \Psi_\theta)}{(\Psi_\theta | \Psi_\theta)} \\ &= \frac{1}{4} \frac{e^{i\theta} \int d\Omega (Y_0^0(\Omega))^2 \int dr r^2 e^{2i\theta} u_\theta(r)^2}{e^{i\theta} \int d\Omega (Y_0^0(\Omega))^2 \int dr u_\theta(r)^2} = \frac{e^{2i\theta}}{4} \int dr r^2 u_\theta(r)^2 \end{aligned}$$

the $1/4$ factor is just to average out the CM. Γ factor is angular contribution (we work exclusively in S wave) and is ignored (though here it cancels out anyway). DVR code is working in relative coordinate of reduced radial equation, so u_ϕ are the eigenvectors.

In DVR quadrature:

$$\langle r_\theta^2 \rangle = e^{2i\theta} \frac{1}{4} \sum_k r_k^2 u_\theta(r_k)^2 \omega_k = e^{2i\theta} \frac{1}{4} \sum_k c_k^2 r_k^2,$$

where $c_k = u_\theta(r_k) \sqrt{\omega_k}$ is the k -th DVR expansion coefficient, and the quadrature weights ω_k cancel due to c-normalization.

5.2.2 Back-rotated radius squared operator

$\langle r^2 \rangle_{\text{BR}}$ is the expectation value of the back-rotated Gamow state. Here r^2 is un-rotated, and $\Psi_{\theta=0} = U_{\theta}^{-1} \Psi_{\theta} = e^{-3i\theta/2} \Psi_{\theta}(re^{-i\phi}) = \frac{u_{\theta}(e^{-i\theta}r)}{r} Y_0^0$. Thus,

$$\begin{aligned} \langle r^2 \rangle_{\text{BR}} &= \frac{1}{4} \frac{\int d\Omega (Y_0^0(\Omega))^2 \int dr r^2 u_{\phi}^2(e^{-i\phi}r)}{\int d\Omega (Y_0^0(\Omega))^2 \int dr u_{\phi}^2(e^{-i\phi}r)} = \frac{1}{4} \frac{\int dr r^2 u_{\phi}^2(e^{-i\phi}r)}{\int dr u_{\phi}^2(e^{-i\phi}r)} \\ &= \frac{1}{4} \frac{e^{i3\theta} \lim_{\epsilon \rightarrow \infty} \int_0^{\epsilon e^{-i\theta}} dr' r'^2 u_{\phi}(r')}{e^{i\theta} \lim_{\epsilon \rightarrow \infty} \int_0^{\epsilon e^{-i\theta}} dr' u_{\phi}(r')} = \frac{e^{2i\theta} \lim_{\epsilon \rightarrow \infty} \int_0^{\epsilon e^{-i\theta}} dr' r'^2 u_{\phi}(r')}{4 \lim_{\epsilon \rightarrow \infty} \int_0^{\epsilon e^{-i\theta}} dr' u_{\phi}(r')}, \end{aligned}$$

where $u_{\phi}(re^{-i\theta}) = u(r)$ is the original Gamow state, and $r' = re^{-i\theta}$. The integral bounds for this new variable are $r' \rightarrow 0$ to $r' \rightarrow \infty$, but, formally, the integral is to be taken along the ray along $e^{-i\theta}$, so $r' \rightarrow 0$ to $r' \rightarrow \infty e^{-i\theta}$. Thus, the ratio above is formally ∞/∞ , but it is well-defined as the boundary value $\theta \rightarrow 0$ of

$$f(\theta) = \frac{e^{2i\theta} \int_0^{\infty} dr u_{\theta}(r)^2 r^2}{\int_0^{\infty} dr u_{\theta}(r)^2}$$

which is constant in the "good" θ sector by the ABC theorem. I.e., this singularity in the function $\langle r_{\theta}^2 \rangle(\theta)$, while not formally defined, approaches a finite limit. Formally, this ratio is only defined in the analytic continuation of this expectation value. Since the ratio is the same constant through the wedge, the analytic continuation to $\theta = 0$ will be the same constant by the identity theorem.

In DVR quadrature,

$$\langle r^2 \rangle_{\text{BR}} = \frac{1}{4} \frac{\sum_k r_k^2 u_{\phi}^2(e^{-i\phi}r_k) \omega_k}{\sum_k u_{\theta}(e^{-i\theta}r_k)^2 \omega_k}.$$

We normalize the states wrt the DVR basis, so the denominator is 1. The back-rotated radius computed in this way is ill-posed as errors accumulate due to back-rotating the decaying state back to the divergent Gamow state.

5.3 Tikhonov Regularization

$$\langle r^2 \rangle_{\text{BR}} = \int dr r^2 u_{\theta}(e^{-i\theta}r) e^{2i\theta}$$

Let $f_{\theta}(r) = r^2 u_{\theta}^2(r)$. Then,

$$\langle r^2 \rangle_{\text{BR}} = e^{4i\theta} \int dr f_{\theta}(e^{-i\theta}r).$$

Changing variables $x = -\ln(r/r_0)$ gives $\tilde{f}_{\theta}(x) = f_{\theta}(r_0 e^{-x})$ (domain goes from $(0, \infty) \rightarrow (-\infty, \infty)$). Fourier transform:

$$\hat{f}_{\theta}(\xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx e^{-ix\xi} \tilde{f}_{\theta}(x)$$

Now, analytically continuing to complex plane: $\tilde{f}_\theta(x + iy) = f_\theta(r_0 e^{-x} e^{-iy})$. So, $\tilde{f}_\theta(x + i\theta) = f_\theta(e^{-i\theta} r)$.

$$\begin{aligned}\tilde{f}_\theta(x + i\theta) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\xi e^{i(x+i\theta)\xi} \hat{f}_\theta(\xi) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\xi e^{ix\xi} e^{-\theta\xi} \hat{f}_\theta(\xi).\end{aligned}$$

Tikhonov cuts off the UV modes: $\Gamma_{\text{Tikhonov}} = \frac{1}{1+\kappa e^{-2\theta\xi}}$. Thus,

$$\tilde{f}_{\text{reg},\theta}(x + i\theta) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\xi e^{ix\xi} e^{-\theta\xi} \hat{f}_\theta(\xi) \Gamma_{\text{Tikhonov}}.$$

Thus,

$$\begin{aligned}\langle r^2 \rangle_{\text{BR}} &= e^{4i\theta} \int_0^\infty dr \tilde{f}_{\text{reg},\theta}(r) = e^{4i\theta} \int_\infty^{-\infty} -r_0 e^{-x} dx \tilde{f}_{\text{reg},\theta}(x + i\theta) \\ &= e^{4i\theta} \int_{-\infty}^\infty r_0 e^{-x} dx \tilde{f}_{\text{reg},\theta}(x + i\theta)\end{aligned}$$

5.4 Zeldovich Regularization

To be added.